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The effect of unsuccessful retrieval on children's subsequent learning

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ABSTRACT

It is well known that successful retrieval enhances subsequent adults' learning by promoting long-term retention. Recent research has also found benefits from unsuccessful retrieval, but the evidence is not as clear-cut when the participants are children. In this study, we employed a methodology based on guessing—the weak associate paradigm—to test whether children can learn from generated errors or whether errors are harmful for learning. We tested second- and third-grade children in Experiment 1 and tested preschool and kindergarten children in Experiment 2. With slight differences in the method, in both experiments children heard the experimenter saying one word (cue) and were asked to guess an associate word (guess condition) or to listen to the correspondent target-associated word (study condition), followed by corrective feedback in both conditions. At the end of the guessing phase, the children undertook a cued-recall task in which they were presented with each cue and were asked to say the corrected target. Together, the results showed that older children—those in kindergarten and early elementary school—benefited from unsuccessful retrieval. Older children showed more correct responses and fewer errors in the guess condition. In contrast, preschoolers produced similar levels of correct and error responses in the two conditions. In conclusion, generating errors seems to be beneficial for future learning of children older than 5 years.

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Introduction

Can we learn from errors? Recent research with adults has shown the benefit of making errors for learning (for a review, see Metcalfe, 2017). However, little is known about how generating errors can affect children's learning. This topic has importance for informing educational practice. Educators may be reluctant to ask children to guess before learning the correct information for fear that their errors would persist and be harmful for future learning. Our study aimed to give an answer to this question and contribute to the debate of whether generating errors is beneficial or detrimental for children's learning.

Effects of successful retrieval on learning in adults and children

Research conducted throughout the last century (e.g., Gates, 1917; Spitzer, 1939; Tulving, 1967) demonstrated that testing plays a more prominent role in learning than simply being a device to assess information. More recently, a revival of interest in this topic replicated and extended previous findings about the benefits of testing for learning. There is now a large body of research showing that retrieval, in the form of tests, enhances learning. For example, Roediger and Karpicke (2006) compared testing previous learning material (retrieval practice) with restudying the information and found that retrieval had a greater benefit for future learning than restudy. This finding—that testing increases long-term retention—has been called the *testing effect* (e.g., Roediger & Karpicke, 2006).

The testing effect has been consistently observed with high school, middle school (e.g., Carpenter, Pashler, & Cepeda, 2009; McDaniel, Agarwal, Huelser, McDermott, & Roediger, 2011), and elementary school children (e.g., Aslan & Bauml, 2015; Bouwmeester & Verkoeijen, 2011; Goossens, Camp, Verkoeijen, & Tabbers, 2014; Karpicke, Blunt, & Smith, 2016; Lipko-Speed, Dunlosky, & Rawson, 2014; Lipowski, Pyc, Dunlosky, & Rawson, 2014; Rohrer, Taylor, & Sholar, 2010). For example, Lipowski et al. (2014) demonstrated the testing effect with 7- and 9-year-old children and found a similar magnitude of the effect between these two age groups. Other authors found that retrieval practice improved vocabulary learning in children aged 8–11 years (Goossens et al., 2014) and helped fifth graders to learn key concepts in science (Lipko-Speed et al., 2014).

The results from the few published studies with younger children are mixed, with one study finding preschoolers' benefit from testing on cued recall (Fritz, Morris, Nolan, & Singleton, 2007) but another showing testing effects only for children above first grade, with younger children not benefiting from it (Gates, 1917).

Is it possible to learn from errors?

Research about the effect of retrieval on subsequent memory has focused on successful retrieval (i.e., the testing effect) and shown that previously testing information enhances its subsequent memory and diminishes the absence of response. However, much less is known about the impact of wrong answers, that is, the influence of the retrieval of commission errors on subsequent memory and, thus, on learning.

When participants first give a wrong answer followed by correct feedback, it remains unclear how that affects subsequent retrievability of the correct answers. Past research on this topic was conducted by Skinner (1968), who introduced a teaching procedure that was designated *errorless learning*, which involves learning without committing any errors or with as few errors as possible. Proponents of errorless learning concluded that this teaching procedure was the best method to promote long-term learning, with several studies showing that there are advantages to errorless learning in individuals with memory impairments or learning disabilities (see Clare & Jones, 2008, for a review). Other researchers have concluded that errorless learning techniques are similarly effective for typical adults (e.g., Warmington & Hitch, 2014) and elementary school children (e.g., Warmington, Hitch, & Gathercole, 2013) on word learning tasks.

The contrasting idea—that retrieval failure can improve learning—was first developed by Izawa (1970). The results of these experiments indicate that retrieval failures seemed to potentiate the amount of learning that occurred on subsequent study trials. More recently, this topic was revived in several studies showing benefits of unsuccessful retrieval across a wide variety of materials (e.g., trivia questions, cue–target pairs). For example, Kornell, Hays, and Bjork (2009, Experiments 3–6) investigated the effects of adults' unsuccessful retrieval in learning using a methodology based on guessing—the weak associate paradigm. In one condition, participants were presented with cues and needed to guess the corresponding targets, followed by correct target feedback (guess condition). In a second condition, participants studied pairs of correct cue–target pairs (study condition). In the guess condition, the correct target was a weak associate, but participants rarely provided it as their response and instead usually generated the strongest associates of the cues. Thus, participants consistently gave responses deemed incorrect. The rare items that participants did successfully guess were excluded from the analyses to ensure that only retrieval failure items were included. During a later cued-recall test phase, participants in both the study and guess conditions were prompted with the cues and asked to recall the correct targets. The results showed that unsuccessful retrieval enhanced learning of the correct targets as compared with the study condition.

Many other studies conducted during the past several years have replicated and extended this finding. One important provision of the effect is that the to-be-learned information needs to be related (e.g., Grimaldi & Karpicke, 2012; Hays, Kornell, & Bjork, 2013; Huelser & Metcalfe, 2012; Knight, Ball, Brewer, Dewitt, & Marsh, 2012), and a second provision is that unsuccessful retrieval needs to be followed by immediate feedback instead of delayed feedback (e.g., Kornell et al., 2009; Vaughn & Rawson, 2012).

An intuitive explanation for this surprising finding is that the discrepancy between the wrong answer and the corrective feedback increases attention to the feedback and, as a result, strengthens learning (e.g., Potts & Shanks, 2014). Specific explanations of the benefits of learning propose that (a) attempting to retrieve information, even incorrect but associated information, may result in deep or elaborated processing within a semantic network, leading to a rich context for encoding the corrective feedback (Bjork, 1975; Kornell et al., 2009); (b) incorrect responses may be inhibited or suppressed after corrective feedback is provided, leading to less interference from the incorrect responses on the final test (e.g., Carrier & Pashler, 1992); and (c) incorrect responses function as mediating cues aiding the recall of the correct target (e.g., Knight et al., 2012; Pyc & Rawson, 2009; see Kornell & Vaughn, 2016, and Yan, Yu, Garcia, & Bjork, 2014, for further discussion).

To date, the question of whether children can benefit from making errors remains unresolved because the results from different studies are not consistent. On the one hand, Warmington et al. (2013) found that eighth-grade children recalled the names of objects more accurately when they were taught using the errorless method than when they were allowed to make errors. On the other hand, Kornell and Metcalfe (2014), using real-world educational materials with middle school children as well as undergraduate students, showed that errors, when followed by feedback, had little or no cost on learning. Moreover, Marsh, Fazio, and Goswick (2012) found that elementary school children can correct commission errors when corrective feedback is provided.

The current study

In the presence of such inconsistent results, and in the absence of studies analyzing the impact of unsuccessful retrieval with early elementary school and preschool-aged children, it is critical to know what the consequences of making errors are for subsequent learning in children at the beginning of their formal education (from preschool and early school grades). Does making errors harm, enhance, or simply have no effect on children's learning?

This issue has important practical implications for educational practice. As far as we know, young children at the beginning of school have not previously been the target of such research. One question is particularly relevant: If a benefit to retrieval does occur, at what age can children take advantage of unsuccessful retrieval and what is its progression across children's age?

Some of the changes in memory over development have implications for what children can learn and for choosing learning methods for children of different ages (for a review, see Bauer & Fivush,

2013). For example, vulnerability to interference tends to decrease as children grow older, from preschool to late childhood (Dempster, 1993; Kail, 2002). This was found both for proactive interference (e.g., in search behavior: Sophian & Wellman, 1983) and for retroactive interference (e.g., in misinformation studies: Ceci & Bruck, 1993). In a meta-analysis based on 28 studies, Kail (2002) found that vulnerability to interference decreased from preschool to late childhood. Moreover, inhibition processes, or the ability to ignore irrelevant information, grow during childhood (e.g., Bjorklund & Harnishfeger, 1990; Dempster & Brainerd, 1995; Harnishfeger & Bjorklund, 1993), as was found in tasks of directed forgetting (Harnishfeger & Pope, 1996). Furthermore, source memory increases with age. Young children demonstrate a poor ability to monitor the sources of their memories (for reviews, see Lindsay, 2002, and Roberts & Blades, 2000).

In the current study, we tested second-grade children in Experiment 1 and tested preschool and kindergarten children in Experiment 2. Considering that elementary school children showed the testing effect (e.g., Lipowski et al., 2014), we generally predicted that older children could use unsuccessful retrieval to enhance learning (compared with study only). In contrast, this would be more demanding for younger children because of their vulnerability to interference, difficulties in inhibiting irrelevant information, and difficulties in discriminating between the wrong generated word and the corrective feedback.

In both experiments, we used the methodology adapted from Kornell et al. (2009) study because it has the advantage of eliciting numerous incorrect responses from participants, which is necessary to study unsuccessful retrieval. However, this method fails to account for differences in proactive interference across conditions: In the study condition that Kornell et al. used, participants did not encounter interference during the experience because they studied only one cue–target association. In contrast, in the guessing condition, the feedback was always different from what had been guessed for the same cue, which could make this condition more difficult to memorize because of interference from the guessed target. To avoid this problem with children, we adapted the methodology of Kornell et al. to accommodate the cognitive limitations of children, as is described in more detail at the beginning of Experiment 1.

In sum, in the guess/retrieval condition in both experiments, children heard the experimenter saying one word (cue) and were asked to guess an associated word (target) for each presented cue.

We expected children of all age groups to produce a strong associate of the cue, as had occurred with adults in the study by Kornell et al. (2009, Experiments 3–6). Immediately after each guess, feedback providing the correct target was presented. The correct target was a weaker associate of the cue and was presented by a cartoon character acting as a teacher correcting the answer (see Fig. 1). The guess answer was considered "unsuccessful retrieval" when participants failed to guess the correct target; however, we are aware that unsuccessful retrieving is not the same as guessing (as is further discussed in the "Results and discussion" section). After all of the cues and feedback were presented, the children took a final cued-recall (cue–?) test on all pairs. The guess condition was compared with two different study conditions (Experiment 1) or with one study condition (Experiment 2). The design of both experiments is presented in Fig. 1.

Experiment 1

This first experiment aimed to replicate and extend Kornell et al. (2009) adult study with children in elementary school. There were three reasons for extending this study. First, we wanted to know whether unsuccessful retrieval would enhance subsequent learning in ages where classroom methods are crucial for learning. Second, we were interested in the correction of errors, that is, whether errors—either generated or heard—would be more easily corrected if the students had previously retrieved them than if they had previously studied them. Third, we aimed to control for potential proactive interference.

Although proactive interference, if it occurred, did not appear to have negative consequences on adults' learning, (e.g., Kornell et al., 2009), it may have more negative impacts on children's learning. In the original study by Kornell et al. (2009), the participants in the study condition were not presented with any incorrect information; that is, participants were exposed only to correct information.

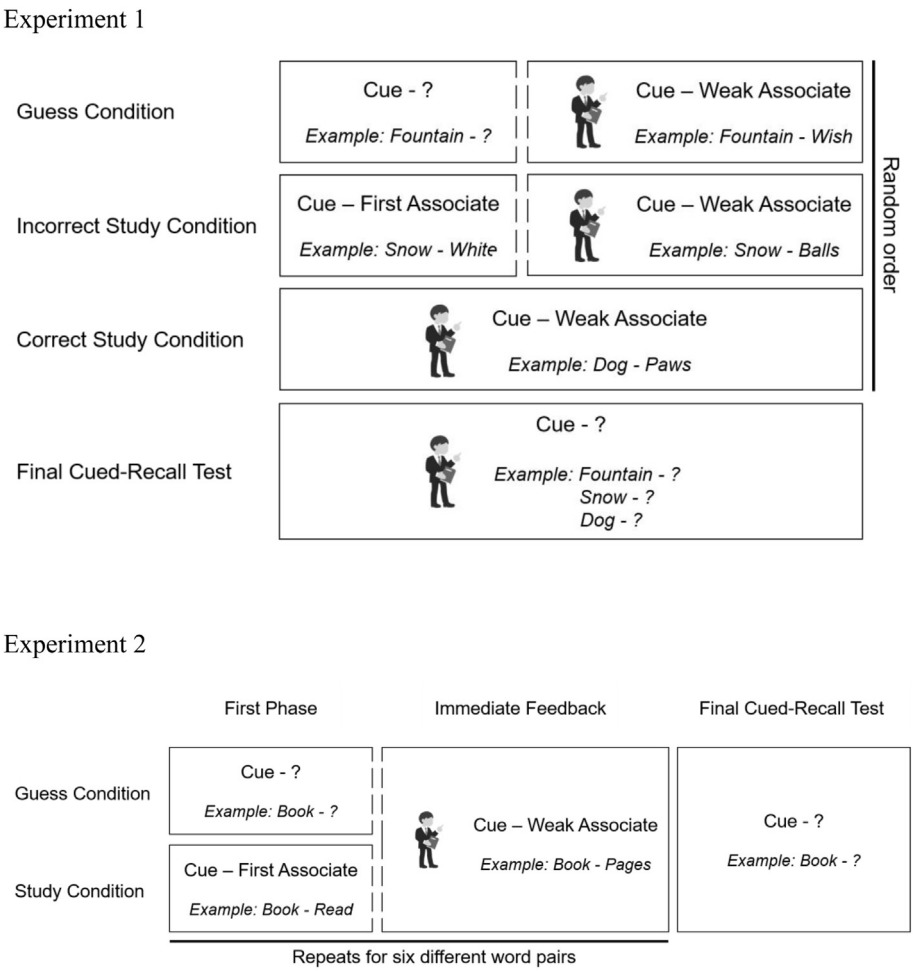


Fig. 1. Procedure of Experiment 1 and Experiment 2.

They always studied the cue and a weak associate (e.g., garden–playground) in the study condition, whereas in the guess condition the generated target (e.g., garden–flowers) was almost always different from the correct target presented in the feedback (e.g., garden–playground). To be successful in the final cued-recall task, participants in the guess condition may have needed to inhibit their previously generated guess response. To address the potential influence of proactive interference from the generated error, in the current experiment we included the previous guess and study conditions and added another study condition in which the same incorrect content generated in the guess condition would be presented before learning the correct target response. In this new study condition (incorrect study condition), children first read a word pair composed of a cue word and a strongly related target (e.g. garden–flowers), which was followed by the corrective feedback (e.g., garden–playground), as in the guess condition. Comparing the guess condition with this incorrect study condition allowed us to analyze the effect of exposure to incorrect information on learning and keep the potential effects of proactive interference constant. Thus, all children performed three conditions: a guess condition in which the guessed target was incorrect and followed by feedback presentation of the correct weak target, an incorrect study condition with the cue–strong target presented first and followed by feedback

presentation of the correct cue–weak target pair, and a correct study condition (similar to Kornell et al.'s (2009) Experiments 3–6) in which participants were not exposed to any incorrect information. The cue and correct weak target were presented for the same amount of time as allotted in the guess plus feedback (guess condition) or studied plus feedback (incorrect study condition). Although we think that it is more appropriate to compare the guess condition with the incorrect study condition, especially with a child population, we decided to also include the correct study condition in our analyses because this study condition is most similar to what actually happens in the real world, where students usually wait for the presentation of the correct answers before memorizing them. Our dependent measure of interest was final cued-recall performance.

We contrasted two competing hypotheses. Elementary school children already show the testing effect (e.g., Lipowski et al., 2014). Thus, one possibility is that the children would also benefit from incorrect retrieval and demonstrate better accuracy and fewer errors on the final cued-recall test in the guess condition as compared with either of the study conditions. Another possibility is that generating or being presented with an incorrect response would impair learning of the correct cue–target association, and performance would be worse in the guess and incorrect study conditions than in the correct study condition in which no incorrect information was ever presented. In the study condition, where potential interference was absent, it is possible that elementary school children would produce more accuracy than in unsuccessful retrieval.

We also evaluated children's confidence in their performance. Based on previous studies that found a tendency for children to be overconfident (Bjorklund, 1997), we expected that children would be overconfident in their responses and show high levels of confidence even when they were wrong.

Method

Participants

A total of 23 Portuguese children in second grade participated in this experiment: 11 7-years-olds and 12 8-years-olds (M_{age} = 7.52 years; 12 female and 11 male). All children were native Portuguese speakers from an elementary school in Lisbon. Children's parents provided written consent for their participation.

Design

This experiment followed a within-participants manipulation design with a task condition independent variable (guess vs. incorrect study vs. correct study) (see Fig. 1). The within-participants manipulation was chosen to eliminate individual differences, following the design usually adopted by the original study by Kornell et al. (2009, except Experiment 6) and studies that analyzed the testing effect with elementary school children (e.g., Bouwmeester & Verkoeijen, 2011; Goossens et al., 2014; Karpicke et al., 2016).

Materials

A total of 30 word pairs following age-specific free association norms for the Portuguese language (Carneiro, Albuquerque, Fernandez, & Esteves, 2004; Comesaña, Fraga, Moreira, Frade, & Soares, 2013) were used in this experiment (see Appendix A).

Each word pair was composed of a cue and either its first associate (considered the strongest associate for the cue) or its eighth associate (considered a weak associate). The forward association strength of the strongly associated word pairs was within a range of .21–.86, meaning that when presented with the cue, 21–86% of participants produced the first associate (Carneiro et al., 2004; Comesaña et al., 2013). The forward association strength of the weakly associated word pairs was within a range of .01–.04, meaning that when presented with the cue, 1–4% of participants produced the eighth associated word. The 30 word pairs were organized into three blocks, one for each condition, composed of 10 word pairs per block. Randomization of word pairs and across conditions was guaranteed.

Procedure

There were three phases during the experiment: an initial phase composed of guess, incorrect study, and correct study tasks; a second phase that included a cued-recall test and an estimation of confidence after each cued-recall test response; and finally a judgment phase in which children indicated which learning condition they had preferred. During the initial phase, each child learned items according to the guess condition, the incorrect study condition, and the correct study condition (performed in a random order). Children were expected to learn 30 word pairs (10 pairs per condition). Afterward, during the final test phase, children took a cued-recall test on all 30 items and made a confidence judgment for each final response. In the end, children were invited to say which of the three conditions they thought were the easiest and hardest learning methods.

At the start of the experiment, children were invited to play a game with the experimenter on a laptop computer in the school library. Before the experiment began, the experimenter explained to the children that they were going to play a game in which they needed to learn and memorize correct word pairs provided by a cartoon character (see Fig. 1). They were told that, at the end of the game, they would be asked to remember the correct word pairs provided by this cartoon character.

In the guess condition, children were asked to read out loud a word cue (e.g., "fountain") presented for 8 s on the computer screen and, during this time, to make a guess out loud as to what the correspondent associated target word was (the answer was registered by the experimenter). It was expected that children would generate a strong associate for the cue (e.g., for the word cue "fountain," it was expected that children would produce a highly associated word such as "water"). Immediately after children generated their guess, the cartoon character appeared on the screen, providing the correct word pair in a written message—composed of the word cue and its weak associate—for 5 s (e.g., "fountain–wish"). Children were asked to read the correct pair out loud and, about every two trials, were reminded that this was the correct information that should be memorized for a later test. This procedure was consecutively repeated 10 times for 10 different word pairs.

In the incorrect study condition, children were asked to read out loud each word pair presented on the computer for 8 s. Children were told that, as part of the game, the computer was trying to guess the word pair. This pair was composed of a word cue and its strongest associate (e.g., "snow–white") to match with the expected guess generated in the guess condition. Immediately after the pair was presented, the cartoon character provided the correct word pair (e.g., "snow–balls") for 5 s. Children were asked to read the correct pair out loud and, about every two trials, were reminded that this was the correct information that should be memorized for a later test. This procedure was consecutively repeated 10 times for 10 different word pairs.

In the correct study condition, children were asked to read out loud a correct word pair provided by the cartoon character (e.g., "dog–paws")—presented for 13 s to match with the total time of the guess condition (8 s of guess + 5 s of feedback)—and were reminded that this was the correct information that should be memorized for a later test. This procedure was consecutively repeated 10 times for 10 different word pairs.

These three conditions (guess, incorrect study, and correct study) were presented in a random order across participants.

To ensure that children understood the procedure and were aware of the correct information to retain throughout the initial phase and across conditions, children were constantly reminded that the correct word pairs they should memorize were the ones provided by the cartoon character. These reminders occurred about every two trials when the cartoon character appeared.

After the conclusion of the initial phase, children were asked to perform a cued-recall test. The cartoon character they had previously seen presented each cue word, and children were asked to recall the correct word (out loud) that the cartoon character had previously provided them. After each response, children were asked how sure they were that the word recalled was the correct target previously provided by the cartoon character. A 3-point scale was presented on the computer screen, and children were asked to indicate whether they felt very sure, slightly sure, or unsure of the accuracy of their response. This procedure was repeated for all 30 word pairs in a random order.

After the final cued-recall test was complete, children were asked to report in which condition they felt the words had been easiest and hardest to learn. To do so, children were presented with pictures of each condition, which were screen shots taken from the initial phase of the experiment, and were

reminded of the initial instructions for each condition. At the end of the experiment, children were thanked for their participation and were given a sticker as a gratification.

Results and discussion

Following the procedure in Kornell et al. (2009, Experiments 3–6), the trials on which participants guessed the correct response or did not guess any word pair were excluded from the analyses. None of the children correctly guessed the correspondent word pair, but 1 participant did not guess any word for two word pairs and 6 participants did not guess any word for one word pair each. As a result, these items were excluded from further analyses, and the final recall of this condition was divided by the proportion of valid guesses. Items in both study conditions—incorrect study and correct study—were not answered at all and, thus, were never excluded.

Comparisons were made among the three conditions on final test performance. We evaluated *correct responses*, *strong associative errors*, *intrusions*, and *omissions* (Fig. 2). Correct responses corresponded to the feedback provided during the initial phase (weak associates given by the cartoon character) or simply presented in the correct study condition. Strong associative errors were responses that corresponded to incorrect guessing (the specific associative errors generated by each child) or studied items of the incorrect study condition (in the correct study condition, errors were not generated or presented during the first phase of the experiment, and so this category of response is absent for this condition). Intrusions correspond to other words recalled in the final cued recall, different from the studied/guessing or the feedback response. Finally, omissions correspond to trials in which children responded "I don't know" in the final cued recall.

Correct responses

A within-participants analysis of variance (ANOVA) comparing the three conditions (guess vs. incorrect study vs. correct study) on proportion correct showed a significant effect of condition, $F(2, 44) = 6.55$, $MSE = .04$, $p = .003$, $\eta_p^2 = .23$. Correct cued recall was higher in the guess condition than in the incorrect study condition ($M_G = .64$, $SD = .19$ vs. $M_{IS} = .43$, $SD = .30$, $p = .01$) and was higher in the guess condition than in the correct study condition ($M_G = .64$, $SD = .18$ vs. $M_{CS} = .47$, $SD = .20$, $p = .02$). The incorrect study and correct study conditions did not differ statistically ($M_{IS} = .43$, $SD = .30$ vs. $M_{CS} = .47$, $SD = .20$, $p = 1.00$). The comparative result between the guess and correct study conditions replicates the adults' results in the Kornell et al. (2009) study, suggesting that unsuccessful retrieval followed by feedback enhanced *correct responses* in the final cued-recall test. The result extends the findings of Kornell et al. with adults to elementary school children.

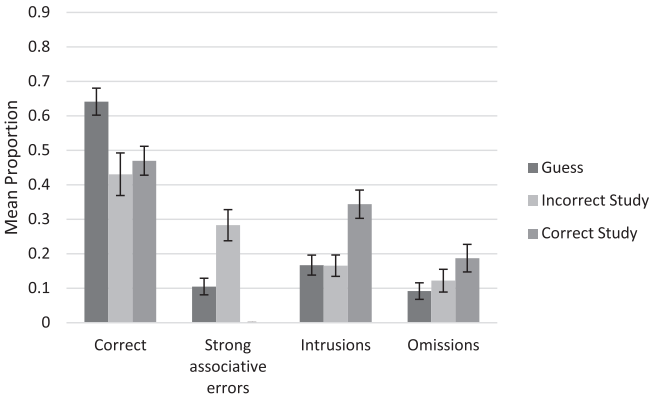


Fig. 2. Mean proportions and standard errors of all four response types (correct responses, strong associative errors, intrusions, and omissions) in guess, incorrect study, and correct study conditions of Experiment 1.

Strong associative errors

In this analysis, we first compared only the guess and incorrect study conditions. The correct study condition was never presented with incorrect target items, and we treated all of the errors on the final cued-recall test in this condition as intrusions. The proportion of strong associative errors was higher in the incorrect study condition than in the guess condition (M_{IS} = .28, SD = .22 vs. M_G = .11, SD = .12), t (22) = 3.58, p = .002, d = 0.76. The findings were novel in that we showed that with elementary school children, fewer of the initial incorrect targets were generated on the final test in the guess condition as compared with a study condition in which incorrect information was also presented.

To complement the analysis, we also compared the proportion of all errors (strong associative errors plus intrusions) among the three conditions. We found a significant effect, F(2, 44) = 6.16, MSE = .034, p = .007, η_p^2 = .22, showing that the strong associate errors plus intrusions in the guess condition produced significantly fewer general errors than incorrect studied errors plus intrusions in the incorrect study condition, (M_G = .27, SD = .20 vs. M_{IS} = .45, SD = .29, p = .02). These errors were also lower in the guess condition compared with the intrusions of the correct study condition, but the difference did not reach significance (M_G = .27, SD = .20 vs. M_{CS} = .34, SD = .21, p = .40). This result suggests that even when we take errors in a broader perspective (strong associative errors plus intrusions), the guess condition also has the advantage in avoiding errors in the face of errors that were previously presented as correct responses. This means that corrective feedback has a more powerful effect when the incorrect information was previously guessed than when it was studied.

Intrusions

A within-participants ANOVA comparing intrusion in each of the three conditions (guess vs. incorrect study vs. correct study) was significant, F(2, 44) = 18.25, MSE = .01, p < .001, η_p^2 = .45, and showed that the rate of intrusions was higher in the correct study condition than in the incorrect study condition (M_{CS} = .34, SD = .21 vs. M_{IS} = .17, SD = .15, p < .001) and was higher in the correct study condition than in the guess condition (M_{CS} = .34, SD = .21 vs. M_G = .17, SD = .14, p = .001). The guess and incorrect study conditions produced similar levels of intrusions (M_G = .17, SD = .14 vs. M_{IS} = .17, SD = .15, p = 1.00). The higher level of intrusions in the correct study condition than in the incorrect study condition was due to the inclusion of all errors in the intrusions category for the correct study condition. Not surprisingly, when incorrect study errors and intrusions were summed and compared with the correct study intrusions, an opposite pattern emerged—more *i*ntrusions plus strong associative errors in the incorrect study condition than in the correct study condition (M_{IS} = .45 vs. M_{CS} = .34).

Omissions

A within-participants ANOVA comparing omissions in each the three conditions (guess vs. incorrect study vs. correct study) was significant, F(2, 44) = 4.41, MSE = .01, p = .02, η_p^2 = .17), showing higher omissions in the correct study condition than in the guess condition (M_{CS} = .19, SD = .19 vs. M_G = .09, SD = .11, p = .04). The omissions between the incorrect study and correct study conditions did not differ statistically (M_{IS} = .12, SD = .16 vs. M_{CS} = .19, SD = .19, p = .14).

Confidence in the responses

Table 1 shows the confidence level in proportion to the correct, errors, and intrusions responses. As can be observed, children seemed to report similar confidence levels across conditions for all response types and, not surprisingly, seemed to report higher confidence levels for correct responses than for strong associative errors or intrusions.

In addition, it is noteworthy that children appear to have chosen the third option, "very sure [of the accuracy of my response]," more frequently—even when they committed errors or made intrusions. In general, this overconfidence of young children in their responses was expected because it was already documented in previous studies (Finn & Metcalfe, 2014; Lipko, Dunlosky, & Merriman, 2009; Shin, Bjorklund, & Beck, 2007).

Retrospective evaluations of learning

At the end of the experiment, when asked to report in which condition they felt it had been easiest to memorize words, 57% of the children chose the guess condition, 19% chose the incorrect study con-

Table 1
Mean proportions (and standard deviations) of confidence responses for correct responses, strong associative errors, and intrusions in guess, incorrect, and correct study conditions.

	Correct				Strong associative errors				Intrusions			
	n	Unsure	Slightly sure	Very sure	*n*	Unsure	Slightly sure	Very sure	*n*	Unsure	Slightly sure	Very sure
Guess	23	.02 (.07)	.06 (.12)	.92 (.16)	14	.11 (.29)	.50 (.62)	.46 (.50)	18	.01 (.06)	.41 (.44)	.58 (.45)
Incorrect study	20	.00 (.00)	.04 (.11)	.95 (.12)	19	.03 (.12)	.31 (.36)	.68 (.33)	17	.06 (.14)	.25 (.37)	.69 (.36)
Correct study	23	.02 (.07)	.12 (.21)	.87 (.21)	–	–	–	–	21	.10 (.17)	.30 (.30)	.60 (.35)

dition, and 24% chose the correct study condition. When asked to report in which condition they felt it was hardest to memorize words, 52% of the children indicated the incorrect study condition, 29% indicated the guess condition, and 19% indicated the correct study condition. These results seem to suggest that children were somewhat aware of the learning improvement produced by the generation of a word compared with mere study and that they were also attentive to the interference that existed in the incorrect study condition.

In sum, the results of this experiment clearly showed a general benefit of unsuccessful retrieval for elementary school children, indicating that retrieval failure compared with incorrect or correct study produced higher accuracy. Moreover, when compared with the incorrect study condition, retrieval failure also produced fewer generated errors on a later cued-recall test, indicating that unsuccessful retrieval leads to better error correction.

Experiment 2

The first experiment demonstrated that 7- and 8-year-old children can learn from errors. Experiment 2 extended Experiment 1 by evaluating whether kindergarten and preschool-aged children would similarly benefit from unsuccessful retrieval and investigating at which age children become able to benefit from it.

Performance on a final cued-recall task was compared between children randomly assigned to the guess condition and those assigned to a study condition. We used only the incorrect study condition because young children are particularly susceptible to proactive interference (Kail, 2002; Yim, Dennis, & Sloutsky, 2013) and also because the correct study condition does not allow the analysis of correction of commission errors. Commission errors were of particular interest because this type of error had never been studied before with this paradigm, representing an advance in relation to the study by Kornell et al. (2009, Experiments 3–6).

We predicted that children aged 4 and 5 years would not benefit from unsuccessful guessing. Contrary to adults and older children, who showed higher final cued-recall performance in the unsuccessful retrieval as compared with the study condition, here we hypothesized that children in both conditions would show similar final test performance. This prediction was based on the findings that young children show greater vulnerability to interference, are sometimes incapable of inhibiting irrelevant information, and have difficulties in monitoring, especially when the sources to be discriminated are very similar semantically (Lindsay, Johnson, & Kwon, 1991). In the current experiment, the two types of word pairs (cue + strong associate guessed or studied and cue + weak associate to be learned) were difficult to discriminate because they are semantically similar. Thus, because of young children's susceptibility to interference from their guesses, and their difficulties in monitoring different sources of information, we expected that in both conditions they would show low levels of correct cued recall.

Regarding the correction of errors, any difference between conditions was predicted to be in the opposite direction from the one found with adults and older children in our study. Knowing that, in general, one's own responses generate stronger memory traces than the heard or read answers of others (the generation effect; Slamecka & Graf, 1978), one possibility was that effects of proactive interference would be greater in the self-generated guess condition than in the study condition for young children. Children might show better memory for the self-generated responses than for the auditorily presented corrective feedback, and because they might fail to discriminate between the two sources of information, it would be expected that they would persist in generating their own incorrect responses. As a result, it was expected that they would show more strong associative errors in the guess condition than in the study condition. This means that the correction of errors with younger children could be more likely in the study condition than in the guess condition.

Method

Participants

A total of 60 Portuguese children participated in this experiment (M_{age} = 4 years 11 months; 25 female and 35 male): 30 preschoolers (M_{age} = 4 years 5 months) and 30 kindergarteners (M_{age} = 5 years

7 months). Half of the participants in both age groups were assigned to the guess condition, and the other half were assigned to the study condition. All children were native Portuguese speakers and were attending preschools in Lisbon. Children's parents provided written consent for their participation.

Design

The independent variable (guess vs. study) was manipulated between participants because we wanted to avoid any confusion that a within-participants design might cause in children of young ages (see Fig. 1).

Materials

A total of 18 word pairs following age-specific free association norms for the Portuguese language (Carneiro et al., 2004) were used in this experiment (see Appendix B). Half of the word pairs were composed of a cue and its first ranked associate (considered the strongest associate for the cue), and the other half were composed of a cue and its seventh- or eighth-ranked associate (considered a weak associate). The forward association strength of the strongly associated word pairs was within a range of .20–.48, meaning that when presented with the cue, 20–48% of participants produced the first associate (Carneiro et al., 2004). The forward association strength of the weakly associated word pairs was within a range of .02–.03, meaning that when presented with the cue, 2–3% of participants produced the seventh or eighth associated word.

The 18 word pairs were organized into three blocks of 6 word pairs each. Each block contained 3 strong associate word pairs and 3 weak associate word pairs. Randomization and counterbalance of word pairs and block presentation was guaranteed.

Procedure

Children were invited to play a game with the experimenter using a laptop computer in a quiet room outside the classroom. Before the experiment began, the experimenter explained to children that they were going to play a game. In the guess condition, they were informed that they were going to try to guess, for each cue word, a corresponding word associated with the cue. In the study condition, they were informed that the experimenter was going to try to guess, for each cue word, a corresponding word associated with the cue. In both conditions, children were told that immediately afterward a cartoon character was going to provide them with the correct word pair and that they should memorize it. Children were also told that later on the experimenter would ask them to remember the correct word pair previously provided by the cartoon character. Children were told what the phrase "word association" meant, were given a few examples, and were asked to do some of them.

During the first phase of the experiment, children in the guess condition heard the experimenter say a cue (e.g., "book") and were asked to guess the correspondent associated word pair. Children in the study condition heard the experimenter say a cue and the correspondent target (it was always the cue's strongest associate to match the expected guess generated in the guess condition (e.g., "b ook–read").

Immediately afterward, in both conditions, the cartoon character appeared on the computer screen and aurally provided the correct correspondent word pair that children should memorize. Children were reminded that this word pair provided by the cartoon character was the actual correct word pair and that they should try to memorize it. In half of the trials, the feedback was composed of cue plus target–weak associate (e.g., "book–pages") to analyze the impact that generating or hearing a wrong response, immediately corrected, has on the final cued-recall test. The other half of the trials of the feedback were composed of cue plus target–first associate (e.g., "book–read") only to motivate the young children for the task, leading them to think that they could actually correctly guess the word pair. However, we did not analyze this last type of trials because they were not of interest for the aim of the current study.

To ensure that children understood the procedure and were aware of the correct information to retain, in all trials during the feedback phase, children were reminded that the correct word pairs that they should memorize were the ones provided by the cartoon character.

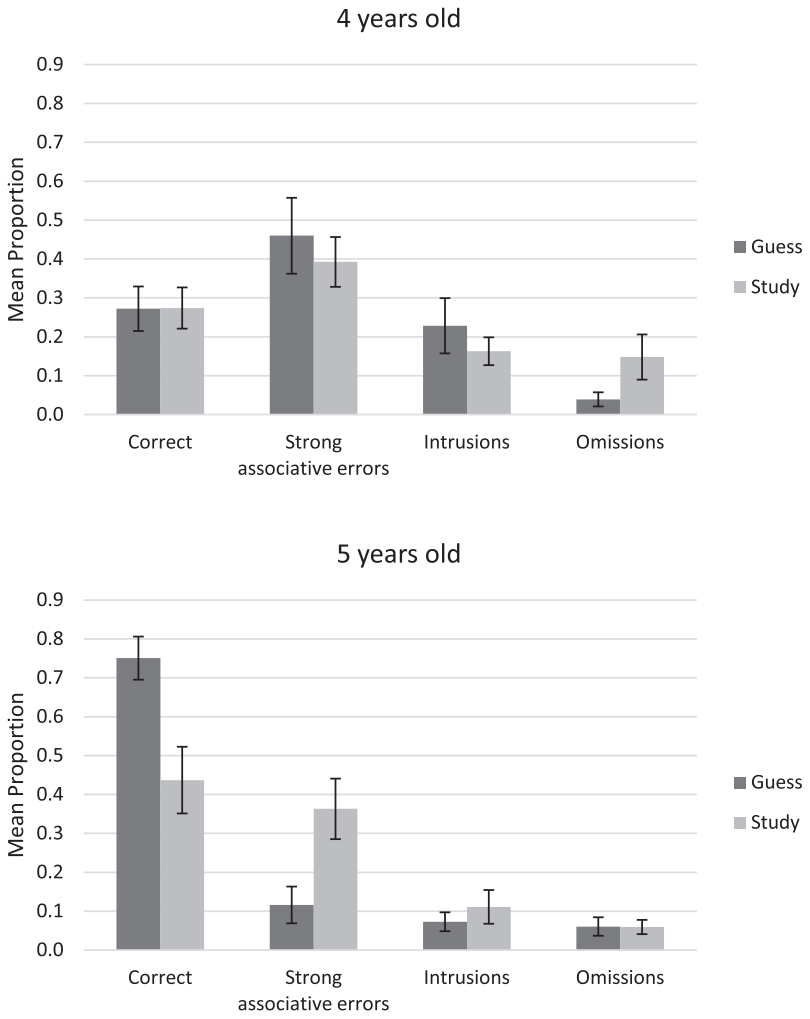


Fig. 3. Mean proportions and standard errors of all four response types (correct responses, strong associative errors, intrusions, and omissions) for children's age in guess and study conditions of Experiment 2.

This procedure was repeated for 6 different trials (3 trials in which the feedback was composed of cue + first associate and 3 trials in which it was composed of cue + weak associate). After 6 trials, children were asked to perform a cued-recall test. The experimenter repeated a cue, and children were asked to say the correct target that had been previously provided to them by the cartoon character (i.e., feedback; e.g., "book - ?"). According to children's willingness to continue, this block (6 trials + cued-recall test) could be repeated three times (for different word pairs), totaling a maximum of 18 trials.

At the end of the experiment, children were thanked for their participation and given a sticker as a gratification.

Results and discussion

During the first phase of the guess condition, 4 participants guessed one correct response and 3 participants did not guess any word for one or two cue words. Thus, these items were excluded in

these participants from further analyses. Final recall performance in this condition was divided by the total number of valid guesses. Items in the study condition were not answered at all during the study phase and, thus, were never excluded.

For the entire sample, we first observed that the guess condition produced higher correct final cued recall than the study condition (M_G = .51, SD = .32 vs. M_S = .36, SD = .28), $t(58) = 1.98$, $p = .05$, $d = 0.50$, and produced similar levels of strong associative errors as compared with the study condition (M_G = .29, SD = .34 vs. M_S = .38, SD = .27), $t(58) = 1.13$, $p = .26$, $d = 0.30$. However, a more interesting result was observed when age groups (preschoolers and kindergarteners) were compared. As can be observed in Fig. 3, a distinct pattern of results was found for each age group. Thus, in the following analyses, we present the comparisons regarding different conditions (retrieval/study) and different age groups.

Four 2 (Age: 4-year-olds vs. 5-year-olds) × 2 (Condition: guess vs. study) between-participants ANOVAs were performed on final performance measures: correct responses, strong associative errors, intrusions, and omissions.

Correct responses

As expected, kindergarteners gave more correct responses than preschoolers in the final cued recall (M_K = .59, SD = .32 vs. M_P = .27, SD = .21), $F(1, 56) = 24.92$, $MSE = .06$, $p < .001$, $\eta_p^2 = .31$. The guess condition produced higher levels of final cued recall than the study condition (M_G = .51, SD = .32 vs. M_S = .36, SD = .28), $F(1, 56) = 5.89$, $MSE = .06$, $p = .02$, $\eta_p^2 = .10$. However, this last result was qualified by a significant Age × Condition interaction, $F(1, 56) = 6.03$, $MSE = .06$, $p = .02$, $\eta_p^2 = .10$, showing that only the 5-year-olds benefited from the guess condition. Preschoolers produced the same level of final correct recall in the guess and study conditions (M_G = .27, SD = .22 vs. M_S = .27, SD = .21), whereas kindergarteners performed much better in the final recall if they previously guessed a strong associate than if they studied the strong associate (M_G = .75, SD = .22 vs. M_S = .44, SD = .33, $p = .001$). Thus, unsuccessful retrieval followed by feedback compared with the study condition plus feedback enhanced learning only for kindergarten children but not for preschool children.

Strong associative errors

As expected, on the final cued-recall test, preschoolers made more errors than kindergarteners (M_P = .43, SD = .32 vs. M_K = .24, SD = .28), $F(1, 56) = 6.37$, $MSE = .08$, $p = .01$, $\eta_p^2 = .10$. In general, there was not a significant effect of condition. However, this last result was qualified by a significant Age × Condition interaction, $F(1, 56) = 4.21$, $MSE = .08$, $p = .04$, $\eta_p^2 = .07$, showing that older children produced more strong associative errors on the final cued-recall test when they studied incorrect information than when they tried to guess it (M_S = .36, SD = .30 vs. M_G = .12, SD = .18, $p = .02$). Younger children produced similar levels of associative errors whenever they guessed (M_G = .46, SD = .38) or studied the pairs (M_S = .39, SD = .25). Moreover, in the guess condition, preschoolers made more errors than kindergarteners (M_P = .46, SD = .38 vs. M_K = .12, SD = .18, $p = .002$). However, in the study condition, the two age groups did not differ significantly (M_P = .39, SD = .25 vs. M_K = .36, SD = .30). Only kindergarteners benefited from the guess condition; retrieval helped them to correct their original guessed errors.

Intrusions

In general, children made few intrusions on the final cued-recall test (M = .14, SD = .19). Preschoolers made more intrusions than kindergarteners (M_P = .20, SD = .22 vs. M_K = .09, SD = .14), $F(1, 56) = 4.90$, $MSE = .03$, $p = .03$, $\eta_p^2 = .08$. No other significant effects were observed.

Omissions

There were no significant effects for the main factors age (M_P = .09, SD = .02 vs. M_K = .06, SD = .02), $F(1, 56) = 0.99$, $MSE = .02$, $p = .32$, $\eta_p^2 = .02$, and condition (M_S = .10, SD = .02 vs. M_G = .05, SD = .02), $F(1, 56) = 2.52$, $MSE = .02$, $p = .12$, $\eta_p^2 = .04$, or for the interaction between the two factors, $F(1, 56) = 2.64$, $MSE = .02$, $p = .11$, $\eta_p^2 = .05$.

In sum, contrary to what was predicted, the results of this experiment showed that kindergarteners were able to benefit from unsuccessful retrieval. This result extends the findings shown with adults (Kornell et al., 2009) and shown with elementary school children in Experiment 1. Kindergarten children, but not preschool-aged children, showed higher correct recall and fewer commission errors

(from either previous guesses or intrusions) on the final cued-recall test in the guess condition than in the study condition. For kindergarteners, correct responses were more easily learned following incorrect retrieval than following study. Preschoolers did not show any advantage of retrieval. The fact that preschoolers gave more correct responses and strong associative errors than intrusions, to some extent, may suggest that these children were not responding by chance. Differences in vulnerability to interference, inhibition, and source memory may underlie the different pattern of results for kindergarten and preschool-aged children. We provide a more extensive discussion of this explanation in the General Discussion.

General discussion

The results of Experiment 1 together with the results of Experiment 2 indicate that children—kindergarten and early elementary school children—benefited from unsuccessful retrieval. Early elementary school children showed that generating a guess followed by corrective feedback was more effective than spending the same time studying information. This was the same pattern that has been shown with adults (Kornell et al., 2009).

The current study included an additional study condition to evaluate whether interference influenced learning of the correct cue–target pair. The conditions in Kornell et al. (2009) study did not equate exposure to incorrect information. For example, in Kornell et al., there was never exposure to incorrect information in the study condition, whereas there was in the guess condition. Here, we extended that paradigm by including a study condition that matched exposure to incorrect information, which allowed a more appropriate comparison with the guess condition. This new study condition had two advantages over the study condition used by Kornell and colleagues. First, in the incorrect study condition, the amount of potential interference between retrieval and study was equated. This is very important when testing children who are more susceptible to the effects of interference than are adults (e.g., Kail, 2002). Second, this condition enabled a comparison between the guess and study conditions of not only correct performance but also the rate of strong associative errors. Note that in the study by Kornell et al., only the final correct responses were analyzed. Our findings are the first to show that unsuccessful retrieval facilitates learning in children from 5 years of age to middle childhood. This finding indicates that the use of guessing, if followed by corrective feedback, is a good method to improve learning in elementary education.

Preschoolers produced similar levels of correct responses and similar levels of strong associative errors in the final cued-recall test in both the guess and study conditions. Thus, unsuccessful retrieval did not have the same effect with preschoolers as it did with kindergarten and elementary school children. This conclusion is in line with studies that investigated children's successful retrieval on learning (i.e., the testing effect) (e.g., Aslan & Bauml, 2015; Gates, 1917). As in the current study, studies of the testing effect with younger children also demonstrate that a development shift seems to occur around these ages.

How can we explain the different pattern shown with kindergarten and preschool-aged children? Although the preschoolers and kindergarteners are not too distant in age, there are major changes in cognitive processing and memory development occurring during this life period (e.g., Lindsay et al., 1991; O'Neill & Gopnik, 1991; Perner, 2000). The development of source memory monitoring abilities that occurs during this age range may provide a potential explanation of the different results obtained in these two age groups (for a review of the studies on children's source memory, see Roberts & Blades, 2000). According to Perner (2000), between 3 and 5 years of age, children acquire an understanding of mind and knowledge, which has implications for the development of memory, especially source memory and resistance to suggestibility. A particular study by O'Neill and Gopnik (1991) demonstrated that changes occur in the ability to identify the source of knowledge between 3 and 5 years of age. These authors showed that 3-year-olds had severe difficulties in identifying the source of their knowledge, whereas 5-year-olds did not show such difficulties. In a memory task, 3-year-olds could remember the objects that were inside a box but could not discriminate whether they knew it because they saw them, because they touched them, or simply because someone told them about the objects. In contrast, 5-year-olds were significantly better in identifying the origins of their memories. Develop-

ment increments in source memory have also been responsible for the increasing immunity to suggestibility across ages (Bruck & Ceci, 1999; Ceci & Bruck, 1993). It is easier to discriminate the false information from correct information if the source of knowledge is identified. The same reasoning can be applied to explain the results of Experiment 2. Distinct effects of guessing on learning for preschoolers and kindergarteners may be due to kindergarteners' ability to better discriminate the two sources of knowledge—what was generated by them and what was corrected by the cartoon character.

Moreover, higher vulnerability to interference (Kail, 2002) and inability to inhibit the self-generated word (Dempster & Brainerd, 1995) can also be responsible for the different results obtained between preschoolers and kindergarteners. Only further studies analyzing unsuccessful retrieval simultaneously with some dimensions of memory development, such as vulnerability to interference, inhibition ability, and source memory, could provide a more valid explanation for such differences.

We worked in this study to provide the preschoolers with careful instructions and certified that they understood that they needed to memorize the corrected words given by the teacher. Nevertheless, what the younger children understood about the task may have been different from what the older children understood. Even so, the main conclusions of the study—that children over 5 years old benefited from unsuccessful retrieval and that preschoolers did not, at least when using the methods employed in our study—seem to be unquestioned. Further exploration of conditions under which young children might benefit from errors is an important avenue for future research.

In general, we observed that early elementary school children tended to be overconfident in their responses. Even on the final test, children were very sure that their incorrect responses were correct 42% of the time in the guess condition and 54% of the time in the incorrect study condition. This result corroborates previous findings indicating that young children are overconfident in their responses (Finn & Metcalfe, 2014; Lipko et al., 2009; Shin et al., 2007). Such general overconfidence in responses did not allow us to conduct an analysis of the hypercorrect effect because responses of low confidence are also needed for comparison. Perhaps with older children, such as in late elementary school, this could be possible. In further studies, it will be interesting to test the confidence of the guess responses during the initial phase of the experiment and to further explore whether more confident guesses lead to increased error correction, as is observed in the hypercorrection effect (Butterfield & Metcalfe, 2001; Metcalfe & Finn, 2012).

There are advantages and disadvantages to using the methodology of the current study—the weak associate paradigm. This methodology has the advantage of allowing a large number of unsuccessful responses to be generated during the first phase of the guess condition. Because associates are easy to generate, participants usually do not leave answers blank, which provides more trials to be analyzed. However, those guess responses cannot be considered right or wrong in the real world. Just in the context of the experiment, we call it a correct guess or a failure. This implies that those responses do not involve an episodic retrieval of a previous context or learning event, which for some authors (e.g., Karpicke, Lehman, & Aue, 2014; Kornell & Vaughn, 2016) is crucial for correctness of wrong answers. Nevertheless, other studies that used other methods involving episodic retrieval and more realistic situations have also obtained similar results with adult populations (e.g., Grimaldi & Karpicke, 2012; Kornell, 2014). The weak associate paradigm is not the only method to study unsuccessful retrieval. Thus, we propose that further studies should be conducted with children using other types of materials that tap into a rich semantic network of possible answers as materials more educationally relevant (e.g., trivia questions adapted for children's age).

In sum, the current study helps to understand whether learning with previous guessing is an appropriate educational method for children. The methods based on guessing the answers—even if they were unsuccessful—are appropriate for enhancing recall of correct answers and for correcting errors in children older than 5 years, but the same cannot be inferred for preschoolers. Educators who are reluctant to use guessing methods or errors in education of children older than 5 years might not need to worry. Unsuccessful retrieval seems to be a useful way of enhancing learning and correcting errors with kindergarten and early elementary school children.

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Appendix A

Word pairs used in Experiment 1 (translated for English) and forward association strength from the word cue to the strongest and weakest associated word pairs.

Word cue	Strongest associate		Weakest associate	
	Item	%	Item	%
Airport	Airplanes	85.71	Runway	1.02
Lightbulb	Light	77.78	On	1.01
Fountain	Water	68.37	Wish	1.02
Blood	Red	57.14	Falls	1.02
Closet	Clothes	54	Furniture	2
Movie theater	Movie	54	Images	2
Motor	Car	53.06	Noise	1.02
Clock	Hours	50.51	Big	2.02
Player	Football	47.96	Football boots	2.04
Snow	White	47.52	Balls	1.98
Rock	Hard	43.88	Small	2.04
Hand	Fingers	43.62	Bread	2.13
Court	Judge	41.86	Chairs	1.16
Heat	Sun	41.41	Good	1.01
Doctor	Physician	41	Medical gown	2
Night	Dark	40	Morning	2
Tree	Leaves	39	Bush	3
Hospital	Patients	38.78	Heal	2.04
Kiss	Love	34.69	Heart	3.06
Iron	Hard	34.69	Wood	2.04
Garden	Flowers	34	Playground	3
Bank	Money	33	Color	2
Climbing	Mountain	27.59	Climber	1.72
Face	Eyes	27.27	Hair	2.02
Edifice	Building	25.88	Bricks	2.35
Mother	Father	24.24	Dear	3.03
Dog	Female dog	23.23	Paws	4.04
Key	Door	22.45	Metal	3.06
Boat	Sea	22.22	Oars	3.03
Bed	Sheets	21.43	Bedspread	3.06

Portuguese word pairs used in Experiment 1 and forward association strength from the word cue to the strongest and weakest associated word pairs.

Word cue	Strongest associate		Weakest associate	
	Item	%	Item	%
Aeroporto	Aviões	85.71	Pista	1.02
Lâmpada	Luz	77.78	Acesa	1.01
Fonte	Água	68.37	Desejo	1.02
Sangue	Vermelho	57.14	Alojar	1.02
Armário	Roupa	54	Móvel	2
Cinema	Filme	54	Imagens	2
Motor	Carro	53.06	Barulho	1.02
Relógio	Horas	50.51	Grande	2.02
Jogador	Futebol	47.96	Chuteiras	2.04
Neve	Branca	47.52	Bolas	1.98
Pedra	Dura	43.88	Pequena	2.04
Mão	Dedos	43.62	Pão	2.13
Tribunal	Juiz	41.86	Cadeiras	1.16
Calor	Sol	41.41	Bom	1.01
Doutor	Médico	41	Bata	2
Noite	Escuro	40	Manhã	2
Árvore	Folhas	39	Arbusto	3
Hospital	Doentes	38.78	Curar	2.04
Beijo	Amor	34.69	Coração	3.06
Ferro	Duro	34.69	Madeira	2.04
Jardim	Flores	34	Parque	3
Banco	Dinheiro	33	Cor	2
Escala	Montanha	27.59	Alpinista	1.72
Cara	Olhos	27.27	Cabelo	2.02
Edifício	Prédio	25.88	Tijolos	2.35
Mãe	Pai	24.24	Querida	3.03
Cão	Cadela	23.23	Patas	4.04
Chave	Porta	22.45	Metal	3.06
Barco	Mar	22.22	Remos	3.03
Cama	Lençóis	21.43	Colcha	3.06

Appendix B

Word pairs used in Experiment 2 (translated for English language) and forward association strength from the word cue to the strongest and weakest associated word pair.

Word cue	Strongest associate		Weakest associate	
	Item	%	Item	%
Eyes	See	48	Games	2
Whine	Drink	42	Cover	2
Door	Open	40	Wood	2
Song	Sing	39	Play	3
Night	Sleep	33	Sun	3

(*continued on next page*)

Appendix B (*continued*)

Word cue	Strongest associate		Weakest associate	
	Item	%	Item	%
Plane	Fly	32	Helicopter	2
Trash	Dustbin	31	Package	2
King	Crown	30	Prince	2
Clothes	Wear	29	Wash	2
Needle	Sew	29	Carpet	2
Car	Wheels	27.27	Light	3.03
Tree	Leaves	27	Street	3
Tooth	Eat	25	Fall	3
Disease	Patient	24	Pain	3
Dog	Woof-Woof	22	Ears	3
Jail	Police	21	Prisoners	3
Book	Read	20	Pages	2
Color	Paint	20	Pink	3

Portuguese word pairs used in Experiment 2 and forward association strength from the word cue to the strongest and weakest associated word pairs.

Word cue	Strongest associate		Weakest associate	
	Item	%	Item	%
Olhos	Ver	48	Jogo	2
Vinho	Beber	42	Tampa	2
Porta	Abrir	40	Madeira	2
Canção	Cantar	39	Brincar	3
Noite	Dormir	33	Sol	3
Avião	Voar	32	Helicóptero	2
Lixo	Caixote	31	Pacote	2
Rei	Coroa	30	Princípe	2
Roupa	Vestir	29	Lavar	2
Agulha	Coser	29	Tapete	2
Carro	Rodas	27.27	Luz	3.03
Árvore	Folhas	27	Rua	3
Dente	Comer	25	Cai	3
Doença	Doente	24	Dor	3
Cão	Ão-Ão	22	Orelhas	3
Prisão	Polícia	21	Prisioneiros	3
Livro	Ler	20	Páginas	2
Cor	Pintar	20	Rosa	3

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